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Formula 106.1:

$$\sum_{k=0}^{n-1}$$

Formula 106.2:

$[x_k, x_{k+1}]$

Formula 106.3:

$$\int_{x_k}^{x_{k+1}} f(x)\,\mathrm{d}x = hf_{k+\frac{1}{2}} + \frac{2}{3!}\left(\frac{h}{2}\right)^3 f''_{k+\frac{1}{2}} + \frac{2}{5!}\left(\frac{h}{2}\right)^5 f^{(4)}_{k+\frac{1}{2}} + \frac{2}{7!}\left(\frac{h}{2}\right)^7 f^{(6)}_{k+\frac{1}{2}} + \cdots,$$

Formula 106.4:

k

Formula 106.5:

0

Formula 106.6:

$n-1$

Formula 106.7:

$$\begin{aligned} I &= \int_a^b f(x)\,\mathrm{d}x \\ &= h\sum f_{k+\frac{1}{2}} + \frac{h^3}{3!\times 2^2}\sum f''_{k+\frac{1}{2}} + \frac{h^5}{5!\times 2^4}\sum f^{(4)}_{k+\frac{1}{2}} + \frac{h^7}{7!\times 2^6}\sum f^{(6)}_{k+\frac{1}{2}} + \cdots. \end{aligned} \tag{4.3.16}$$

Formula 106.8:

$h\sum f_{k+\frac{1}{2}}$

Formula 106.9:

$$T(h) = I + \frac{h^2}{2!\times 6}\sum f''_{k+\frac{1}{2}} + \frac{h^5}{4!\times 20}\sum f^{(4)}_{k+\frac{1}{2}} + \frac{3h^7}{6!\times 224}\sum f^{(6)}_{k+\frac{1}{2}} + \cdots. \tag{4.3.17}$$

Formula 106.10:

$f''(x)$

Formula 106.11:

$$\int_a^b f''(x)\,\mathrm{d}x = f'(b) - f'(a),$$

Formula 106.12:

$$h\sum f''_{k+\frac{1}{2}} = f'(b) - f'(a) - \frac{h^3}{3!\times 2^2}\sum f^{(4)}_{k+\frac{1}{2}} - \frac{h^5}{5!\times 2^4}\sum f^{(6)}_{k+\frac{1}{2}} + \cdots,$$

Formula 106.13:

$$T(h) = I + \frac{h^2}{2!\times 6}[f'(b) - f'(a)] - \frac{h^5}{4!\times 30}\sum f^{(4)}_{k+\frac{1}{2}} - \frac{h^7}{6!\times 56}\sum f^{(6)}_{k+\frac{1}{2}} + \cdots.$$

Formula 106.14:

$f^{(4)}(x)$

Formula 106.15:

$$h\sum f^{(4)}_{k+\frac{1}{2}} = f'''(b) - f'''(a) - \frac{h^3}{3!\times 2^2}\sum f^{(6)}_{k+\frac{1}{2}} + \cdots,$$

Formula 106.16:

$$
\begin{aligned}
T(h) = I &+ \frac{h^2}{2! \times 6}[f'(b) - f'(a)] - \frac{h^4}{4! \times 30}[f'''(b) - f'''(a)] \\
&+ \frac{h^7}{6! \times 42} \sum f^{(6)}_{k+\frac{1}{2}} - \cdots .
\end{aligned} \tag{4.3.18}
$$

Formula 106.17:

h^2

Formula 106.18:

f''

Formula 106.19:

$h^3/12 = h^3/(2! \times 6)$

Formula 106.20:

$h \sum f''$

Formula 106.21:

$h^2[f'(b) - f'(a)]/12$

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Formula 107.1:

$$\int_a^b f(x)\,\mathrm{d}x \approx \sum_{k=0}^{n} A_k f(x_k) \tag{4.4.1}$$

Formula 107.2:
$2n+2$
Formula 107.3:
x_k, A_k
Formula 107.4:
$k = 0, 1, \cdots, n$
Formula 107.5:
$2n+1$
Formula 107.6:
$2n+1$
Formula 107.7:
x_k
Formula 107.8:
$k = 0, 1, \cdots, n$
Formula 107.9:
x_k
Formula 107.10:
$k = 0, 1, \cdots, n$
Formula 107.11:
$\omega(x) = \prod_{k=0}^{n}(x - x_k)$
Formula 107.12:
n
Formula 107.13:
$P(x)$
Formula 107.14:

$$\int_a^b P(x)\omega(x)\,\mathrm{d}x = 0. \tag{4.4.2}$$

Formula 107.15:
$P(x)$
Formula 107.16:
n
Formula 107.17:
$P(x)\omega(x)$
Formula 107.18:
$2n+1$
Formula 107.19:
$x_0, x_1, \cdots, x_n$
Formula 107.20:
$P(x)\omega(x)$
Formula 107.21:

$$\int_a^b P(x)\omega(x)\,\mathrm{d}x = \sum_{k=0}^{n} A_k P(x_k)\omega(x_k).$$

Formula 107.22:

$\omega(x_k) = 0$
Formula 107.23:
$k = 0, 1, \cdots, n$
Formula 107.24:
$2n + 1$
Formula 107.25:
$f(x)$
Formula 107.26:
$\omega(x)$
Formula 107.27:
$f(x)$
Formula 107.28:
$P(x)$
Formula 107.29:
$Q(x)$
Formula 107.30:
$P(x)$
Formula 107.31:
$Q(x)$
Formula 107.32:
n
Formula 107.33:

$$f(x) = P(x)\omega(x) + Q(x).$$

Formula 107.34:

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b Q(x)\,\mathrm{d}x. \tag{4.4.3}$$

Formula 107.35:
$Q(x)$
Formula 107.36:

$$\int_a^b Q(x)\,\mathrm{d}x = \sum_{k=0}^{n} A_k Q(x_k).$$

Formula 107.37:
$\omega(x_k) = 0$
Formula 107.38:
$Q(x_k) = f(x_k)$
Formula 107.39:

$$\int_a^b Q(x)\,\mathrm{d}x = \sum_{k=0}^{n} A_k f(x_k),$$

Formula 107.40:

$$\int_a^b f(x)\,\mathrm{d}x = \sum_{k=0}^{n} A_k f(x_k).$$

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Formula 108.1:
$2n+1$
Formula 108.2:
x_k
Formula 108.3:
$k=0,1,\cdots,n$
Formula 108.4:
$a=-1,b=1$
Formula 108.5:
$[-1,1]$
Formula 108.6:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx \sum_{k=0}^{n} A_k f(x_k). \tag{4.4.4}$$

Formula 108.7:
$[-1,1]$
Formula 108.8:
$P_{n+1}(x)$
Formula 108.9:
$P_1(x)=x$
Formula 108.10:
$x_0=0$
Formula 108.11:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx A_0 f(0),$$

Formula 108.12:
$f(x)=1$
Formula 108.13:
$A_0=2$
Formula 108.14:
$P_2(x)=\frac{1}{2}(3x^2-1)$
Formula 108.15:
$\pm\frac{1}{\sqrt{3}}$
Formula 108.16:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx A_0 f\left(-\frac{1}{\sqrt{3}}\right) + A_1 f\left(\frac{1}{\sqrt{3}}\right),$$

Formula 108.17:
$f(x)=1,x$
Formula 108.18:

$$\begin{cases} A_0 + A_1 = 2, \\ A_0\left(-\frac{1}{\sqrt{3}}\right) + A_1\left(\frac{1}{\sqrt{3}}\right) = 0. \end{cases}$$

Formula 108.19:
$A_0=A_1=1$
Formula 108.20:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right).$$

Formula 108.21:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x$$
$$\approx \frac{5}{9} f\left(-\frac{\sqrt{15}}{5}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\frac{\sqrt{15}}{5}\right).$$

Formula 108.22:
n
Formula 108.23:
x_k
Formula 108.24:
A_k
Formula 108.25:
$0.000\,000\,0$
Formula 108.26:
$2.000\,000\,0$
Formula 108.27:
$\pm 0.577\,350\,3$
Formula 108.28:
$1.000\,000\,0$
Formula 108.29:
$\pm 0.774\,596\,7$
Formula 108.30:
$0.555\,555\,6$
Formula 108.31:
$0.000\,000\,0$
Formula 108.32:
$0.888\,888\,9$
Formula 108.33:
$\pm 0.861\,136\,3$
Formula 108.34:
$0.347\,854\,8$
Formula 108.35:
$\pm 0.339\,881\,0$
Formula 108.36:
$0.652\,145\,2$
Formula 108.37:
$\pm 0.906\,179\,3$
Formula 108.38:
$0.236\,926\,9$
Formula 108.39:
$\pm 0.538\,469\,3$
Formula 108.40:
$0.478\,628\,7$
Formula 108.41:
$0.000\,000\,0$
Formula 108.42:
$0.568\,888\,9$
Formula 108.43:

$[a, b]$
Formula 108.44:
$x = \frac{b-a}{2} t + \frac{a+b}{2}$
Formula 108.45:
$[-1, 1]$
Formula 108.46:

$$\int_a^b f(x)\,\mathrm{d}x = \frac{b-a}{2} \int_{-1}^{1} f\left(\frac{b-a}{2} t + \frac{a+b}{2}\right)\,\mathrm{d}t.$$

Formula 108.47:
$n = 3$
Formula 108.48:
$\pm 0.339\,881\,0$
Formula 108.49:
$P_4(x) = (35x^4 - 30x^2 + 3)/8$
Formula 108.50:
$\sqrt{(15 - 2\sqrt{30})/35} \approx 0.339\,981\,0436$
Formula 108.51:
$0.339\,981\,0$
Formula 108.52:
$0.652\,145\,2$

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Formula 109.1:

$$R(x) = \int_a^b f(x)\,\mathrm{d}x - \sum_{k=0}^{n} A_k f(x_k) = \frac{f^{(2n+2)}(\xi)}{(2n+2)!}\int_a^b \omega^2(x)\,\mathrm{d}x, \tag{4.4.5}$$

Formula 109.2:
$\omega(x) = (x-x_0)(x-x_1)\cdots(x-x_n)$
Formula 109.3:
$x_0, x_1, \cdots, x_n$
Formula 109.4:
$2n+1$
Formula 109.5:
$H(x)$
Formula 109.6:

$$H(x_i) = f(x_i), \quad H'(x_i) = f'(x_i) \qquad (i = 0, 1, \cdots, n).$$

Formula 109.7:
$H(x)$
Formula 109.8:
$2n+1$
Formula 109.9:
$H(x)$
Formula 109.10:

$$\int_a^b H(x)\,\mathrm{d}x = \sum_{k=0}^{n} A_k H(x_k) = \sum_{k=0}^{n} A_k f(x_k),$$

Formula 109.11:

$$\begin{aligned} R(x) &= \int_a^b f(x)\,\mathrm{d}x - \sum_{k=0}^{n} A_k f(x_k) = \int_a^b f(x)\,\mathrm{d}x - \int_a^b H(x)\,\mathrm{d}x \\ &= \int_a^b \frac{f^{(2n+2)}(\eta)}{(2n+2)!}\omega^2(x)\,\mathrm{d}x. \end{aligned}$$

Formula 109.12:
$\omega^2(x)$
Formula 109.13:
$[a, b]$
Formula 109.14:
A_k
Formula 109.15:
$k = 0, 1, \cdots, n$
Formula 109.16:

$$l_k(x) = \prod_{\substack{j=0 \\ j\neq k}}^{n} \frac{x - x_j}{x_k - x_j},$$

Formula 109.17:
n
Formula 109.18:
$l_k^2(x)$

Formula 109.19:
$2n$
Formula 109.20:

$$\int_a^b l_k^2(x)\,\mathrm{d}x = \sum_{i=0}^{n} A_i l_k^2(x_i).$$

Formula 109.21:
$l_k(x_i) = \delta_{ki}$
Formula 109.22:
A_k
Formula 109.23:
$A_k = \int_a^b l_k^2(x)\,\mathrm{d}x > 0$
Formula 109.24:
$I_n = \sum_{k=0}^{n} A_k f(x_k)$
Formula 109.25:
$f_k = f(x_k)$

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Formula 110.1:
f_k^*
Formula 110.2:

$$I_n^* = \sum_{k=0}^{n} A_k f_k^*.$$

Formula 110.3:

$$|I_n^* - I_n| \le \sum_{k=0}^{n} A_k |f_k^* - f_k| \le \left(\sum_{k=0}^{n} A_k\right) \max_{0\le k\le n} |f_k^* - f_k|.$$

Formula 110.4:

$$|I_n^* - I_n| \le (b-a) \max_{0\le k\le n} |f_k^* - f_k|,$$

Formula 110.5:
$I = \int_a^b \rho(x) f(x)\,\mathrm{d}x$
Formula 110.6:
$\rho(x) \ge 0$
Formula 110.7:
$\rho(x) \equiv 1$
Formula 110.8:

$$\int_a^b \rho(x) f(x)\,\mathrm{d}x \approx \sum_{k=0}^{n} A_k f(x_k),$$

Formula 110.9:
$2n+1$
Formula 110.10:
x_k
Formula 110.11:
x_k
Formula 110.12:
$k = 0, 1, \cdots, n$
Formula 110.13:
$\omega(x) = \prod_{k=0}^{n} (x - x_k)$
Formula 110.14:
$[a, b]$
Formula 110.15:
$\rho(x)$
Formula 110.16:
$a = -1, b = 1$
Formula 110.17:
$\rho(x) = \frac{1}{\sqrt{1-x^2}}$
Formula 110.18:

$$\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\,\mathrm{d}x \approx \sum_{k=0}^{n} A_k f(x_k). \tag{4.4.6}$$

Formula 110.19:
$[-1,1]$
Formula 110.20:
$\frac{1}{\sqrt{1-x^2}}$
Formula 110.21:
$n+1$
Formula 110.22:

$$x_k = \cos\left(\frac{2k+1}{2n+2}\pi\right) \qquad (k=0,1,\cdots,n).$$

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Formula 111.1:

$$\int_0^1 \sqrt{x} f(x)\,\mathrm{d}x \approx A_0 f(x_0) + A_1 f(x_1), \tag{4.4.7}$$

Formula 111.2:
$f(x) = 1, x, x^2, x^3$
Formula 111.3:

$$\begin{cases} A_0 + A_1 = \dfrac{2}{3}, \\ x_0 A_0 + x_1 A_1 = \dfrac{2}{5}, \\ x_0^2 A_0 + x_1^2 A_1 = \dfrac{2}{7}, \\ x_0^3 A_0 + x_1^3 A_1 = \dfrac{2}{9}. \end{cases} \tag{4.4.8}$$

Formula 111.4:

$$x_0 A_0 + x_1 A_1 = x_0(A_0 + A_1) + (x_1 - x_0)A_1,$$

Formula 111.5:

$$\frac{2}{3}x_0 + (x_1 - x_0)A_1 = \frac{2}{5}.$$

Formula 111.6:

$$\frac{2}{5}x_0 + (x_1 - x_0)x_1 A_1 = \frac{2}{7}, \qquad \frac{2}{7}x_0 + (x_1 - x_0)x_1^2 A_1 = \frac{2}{9}.$$

Formula 111.7:
$(x_1 - x_0)A_1$
Formula 111.8:

$$\begin{cases} \dfrac{2}{5}x_0 + \left(\dfrac{2}{5} - \dfrac{2}{3}x_0\right) x_1 = \dfrac{2}{7}, \\ \dfrac{2}{7}x_0 + \left(\dfrac{2}{7} - \dfrac{2}{5}x_0\right) x_1^2 = \dfrac{2}{9}, \end{cases}$$

Formula 111.9:

$$\begin{cases} \dfrac{2}{5}(x_0 + x_1) - \dfrac{2}{3}x_0 x_1 = \dfrac{2}{7}, \\ \dfrac{2}{7}(x_0 + x_1) - \dfrac{2}{5}x_0 x_1 = \dfrac{2}{9}. \end{cases}$$

Formula 111.10:

$$x_0 x_1 = \frac{5}{21}, \quad x_0 + x_1 = \frac{10}{9},$$

Formula 111.11:

$$\begin{aligned} x_0 &= 0.821\,162, \quad x_1 = 0.289\,949, \\ A_0 &= 0.389\,111, \quad A_1 = 0.277\,556. \end{aligned}$$

Formula 111.12:

$$\int_0^1 \sqrt{x} f(x)\,\mathrm{d}x \approx 0.389\,111 f(0.821\,162) + 0.277\,556 f(0.289\,949).$$

Formula 111.13:

$$\frac{2}{7}x_0 + \left(\frac{2}{7} - \frac{2}{5}x_0\right) x_1^2 = \frac{2}{9}$$

Formula 111.14:

$$x_1^2$$

Formula 111.15:

$$(x_1 - x_0)x_1 A_1 = \frac{2}{7} - \frac{2}{5}x_0$$

Formula 111.16:

$$x_1$$

Formula 111.17:

$$\left(\frac{2}{7} - \frac{2}{5}x_0\right) x_1$$

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Formula 112.1:
$f'(a)$
Formula 112.2:
$\frac{f(a+h)-f(a)}{h}$
Formula 112.3:
$h \to 0$
Formula 112.4:

$$f'(a) \approx \frac{f(a+h)-f(a)}{h}.$$

Formula 112.5:

$$f'(a) \approx \frac{f(a)-f(a-h)}{h}$$

Formula 112.6:

$$f'(a) \approx \frac{f(a+h)-f(a-h)}{2h}.$$

Formula 112.7:
AB
Formula 112.8:
AC
Formula 112.9:
BC
Formula 112.10:
AT
Formula 112.11:
$f'(a)$
Formula 112.12:
BC
Formula 112.13:
AT
Formula 112.14:
x
Formula 112.15:
y
Formula 112.16:
O
Formula 112.17:
$a-h$
Formula 112.18:
a
Formula 112.19:
$a+h$
Formula 112.20:
C
Formula 112.21:
A
Formula 112.22:
B
Formula 112.23:

$f(x)$
Formula 112.24:
A
Formula 112.25:
T
Formula 112.26:
AB
Formula 112.27:
AC
Formula 112.28:
BC
Formula 112.29:
AT
Formula 112.30:
f
Formula 112.31:

$$G(h) = \frac{f(a+h) - f(a-h)}{2h}$$

Formula 112.32:
$f'(a)$
Formula 112.33:
$f(a \pm h)$
Formula 112.34:
$x = a$
Formula 112.35:

$$f(a \pm h) = f(a) \pm hf'(a) + \frac{h^2}{2!}f''(a) \pm \frac{h^3}{3!}f'''(a) + \frac{h^4}{4!}f^{(4)}(a) \pm \frac{h^5}{5!}f^{(5)}(a) + \cdots ,$$

Formula 112.36:

$$G(h) = f'(a) + \frac{h^2}{3!}f'''(a) + \frac{h^4}{5!}f^{(5)}(a) + \cdots .$$

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Formula 113.1:
h
Formula 113.2:
$f(a+h)$
Formula 113.3:
$f(a-h)$
Formula 113.4:
$f(x)=\sqrt{x}$
Formula 113.5:
$x=2$
Formula 113.6:

$$G(h)=\frac{\sqrt{2+h}-\sqrt{2-h}}{2h}.$$

Formula 113.7:
$f'(2)=0.353\,553$
Formula 113.8:
h
Formula 113.9:
$G(h)$
Formula 113.10:
h
Formula 113.11:
$G(h)$
Formula 113.12:
h
Formula 113.13:
$G(h)$
Formula 113.14:
1
Formula 113.15:
$0.366\,0$
Formula 113.16:
0.05
Formula 113.17:
$0.353\,0$
Formula 113.18:
0.001
Formula 113.19:
$0.350\,0$
Formula 113.20:
0.5
Formula 113.21:
$0.356\,4$
Formula 113.22:
0.01
Formula 113.23:
$0.350\,0$
Formula 113.24:
$0.000\,5$
Formula 113.25:

$0.300\,0$
Formula 113.26:
0.1
Formula 113.27:
$0.353\,5$
Formula 113.28:
0.005
Formula 113.29:
$0.350\,0$
Formula 113.30:
$0.000\,1$
Formula 113.31:
$0.300\,0$
Formula 113.32:
$h = 0.1$
Formula 113.33:
$y = f(x)$
Formula 113.34:
$y = P_n(x)$
Formula 113.35:
$P_n'(x)$
Formula 113.36:
$f'(x)$
Formula 113.37:

$$f'(x) \approx P_n'(x) \tag{4.5.1}$$

Formula 113.38:
x
Formula 113.39:
x_0
Formula 113.40:
x_1
Formula 113.41:
x_2
Formula 113.42:
$\dots$
Formula 113.43:
x_n
Formula 113.44:
y
Formula 113.45:
y_0
Formula 113.46:
y_1
Formula 113.47:
y_2
Formula 113.48:
$\dots$
Formula 113.49:
y_n
Formula 113.50:
$f(x)$

Formula 113.51:
$P_n(x)$
Formula 113.52:
$P_n'(x)$
Formula 113.53:
$f'(x)$
Formula 113.54:

$$f'(x) - P_n'(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\omega_{n+1}'(x) + \frac{\omega_{n+1}(x)}{(n+1)!}\frac{\mathrm{d}}{\mathrm{d}x}f^{(n+1)}(\xi),$$

Formula 113.55:

$$\omega_{n+1}(x) = \prod_{k=0}^{n}(x - x_i).$$

Formula 113.56:
ξ
Formula 113.57:
x
Formula 113.58:
$\frac{\omega_{n+1}(x)}{(n+1)!}\frac{\mathrm{d}}{\mathrm{d}x}f^{(n+1)}(\xi)$
Formula 113.59:
x
Formula 113.60:
$f'(x) - P_n'(x)$
Formula 113.61:
x_k
Formula 113.62:
$\omega_{n+1}(x_k) = 0$
Formula 113.63:
$\prod_{k=0}^{n}(x - x_i)$
Formula 113.64:
k
Formula 113.65:
i
Formula 113.66:
$\omega_{n+1}(x_k) = 0$
Formula 113.67:
$(x - x_k)$
Formula 113.68:
i

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Formula 114.1:

$$f'(x_k) - P_n'(x_k) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\omega_{n+1}'(x_k). \tag{4.5.2}$$

Formula 114.2:
x_0, x_1
Formula 114.3:
$f(x_0), f(x_1)$
Formula 114.4:

$$P_1(x) = \frac{x - x_1}{x_0 - x_1} f(x_0) + \frac{x - x_0}{x_1 - x_0} f(x_1).$$

Formula 114.5:
$x_1 - x_0 = h$
Formula 114.6:

$$P_1'(x) = \frac{1}{h}[-f(x_0) + f(x_1)],$$

Formula 114.7:

$$P_1'(x_0) = \frac{1}{h}[f(x_1) - f(x_0)], \quad P_1'(x_1) = \frac{1}{h}[f(x_1) - f(x_0)].$$

Formula 114.8:

$$f'(x_0) = \frac{1}{h}[f(x_1) - f(x_0)] - \frac{h}{2} f''(\xi),$$

Formula 114.9:

$$f'(x_1) = \frac{1}{h}[f(x_1) - f(x_0)] + \frac{h}{2} f''(\xi).$$

Formula 114.10:
$x_0, x_1 = x_0 + h, x_2 = x_0 + 2h$
Formula 114.11:

$$\begin{aligned} P_2(x) = & \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) \\ & + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2), \end{aligned}$$

Formula 114.12:
$x = x_0 + th$
Formula 114.13:

$$P_2(x_0 + th) = \frac{1}{2}(t-1)(t-2)f(x_0) - t(t-2)f(x_1) + \frac{1}{2}t(t-1)f(x_2),$$

Formula 114.14:
t
Formula 114.15:

$$P_2'(x_0 + th) = \frac{1}{2h}[(2t-3)f(x_0) - (4t-4)f(x_1) + (2t-1)f(x_2)]. \tag{4.5.3}$$

Formula 114.16:
x

Formula 114.17:
$t = 0, 1, 2$
Formula 114.18:

$$P_2'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_1) - f(x_2)],$$

Formula 114.19:

$$P_2'(x_1) = \frac{1}{2h}[-f(x_0) + f(x_2)],$$

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Formula 115.1:

$$P_2'(x_2) = \frac{1}{2h}[f(x_0) - 4f(x_1) + 3f(x_2)],$$

Formula 115.2:

$$f'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_1) - f(x_2)] + \frac{h^2}{3}f'''(\xi),$$

Formula 115.3:

$$f'(x_1) = \frac{1}{2h}[-f(x_0) + f(x_2)] - \frac{h^2}{6}f'''(\xi), \tag{4.5.4}$$

Formula 115.4:

$$f'(x_2) = \frac{1}{2h}[f(x_0) - 4f(x_1) + 3f(x_2)] + \frac{h^2}{3}f'''(\xi),$$

Formula 115.5:
$f(x_1)$
Formula 115.6:
$P_n(x)$
Formula 115.7:
$f(x)$
Formula 115.8:

$$f^{(k)}(x) \approx P_n^{(k)}(x), \quad k = 1, 2, \cdots .$$

Formula 115.9:
t
Formula 115.10:

$$P_2''(x_0 + th) = \frac{1}{h^2}[f(x_0) - 2f(x_1) + f(x_2)],$$

Formula 115.11:

$$P_2''(x_1) = \frac{1}{h^2}[f(x_1 - h) - 2f(x_1) + f(x_1 + h)],$$

Formula 115.12:

$$f''(x_1) = \frac{1}{h^2}[f(x_1 - h) - 2f(x_1) + f(x_1 + h)] - \frac{h^2}{12}f^{(4)}(\xi). \tag{4.5.5}$$

Formula 115.13:
$x_i = x_0 + ih, \quad i = 0, 1, 2, 3, 4$
Formula 115.14:

$$m_0 = \frac{1}{12h}[-25f(x_0) + 48f(x_1) - 36f(x_2) + 16f(x_3) - 3f(x_4)],$$

Formula 115.15:

$$m_1 = \frac{1}{12h}[-3f(x_0) - 10f(x_1) + 18f(x_2) - 6f(x_3) + f(x_4)],$$

Formula 115.16:

$$m_2 = \frac{1}{12h}[f(x_0) - 8f(x_1) + 8f(x_3) - f(x_4)],$$

Formula 115.17:

$$m_3 = \frac{1}{12h}[-f(x_0) + 6f(x_1) - 18f(x_2) + 10f(x_3) + 3f(x_4)],$$

Formula 115.18:

$$m_4 = \frac{1}{12h}[3f(x_0) - 16f(x_1) + 36f(x_2) - 48f(x_3) + 25f(x_4)].$$

Formula 115.19:

m_i

Formula 115.20:

$f'(x_i)$

Formula 115.21:

M_i

Formula 115.22:

$f''(x_i)$

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Formula 116.1:

$$M_0 = \frac{1}{12h^2}[35f(x_0) - 104f(x_1) + 114f(x_2) - 56f(x_3) + 11f(x_4)],$$

Formula 116.2:

$$M_1 = \frac{1}{12h^2}[11f(x_0) - 20f(x_1) + 6f(x_2) + 4f(x_3) - f(x_4)],$$

Formula 116.3:

$$M_2 = \frac{1}{12h^2}[-f(x_0) + 16f(x_1) - 30f(x_2) + 16f(x_3) - f(x_4)],$$

Formula 116.4:

$$M_3 = \frac{1}{12h^2}[-f(x_0) + 4f(x_1) + 6f(x_2) - 20f(x_3) + 11f(x_4)],$$

Formula 116.5:

$$M_4 = \frac{1}{12h^2}[11f(x_0) - 56f(x_1) + 114f(x_2) - 104f(x_3) + 35f(x_4)].$$

Formula 116.6:
$f(x) = \sqrt{x}$
Formula 116.7:
m_i
Formula 116.8:
M_i
Formula 116.9:
x_i
Formula 116.10:
$f(x_i)$
Formula 116.11:
m_i
Formula 116.12:
$f'(x_i)$
Formula 116.13:
$M_i / \times 10^3$
Formula 116.14:
$f''(x_i) / \times 10^3$
Formula 116.15:
100
Formula 116.16:
$10.000\,000$
Formula 116.17:
$0.050\,000$
Formula 116.18:
$0.050\,000$
Formula 116.19:
$-0.247\,58$
Formula 116.20:
$-0.250\,00$
Formula 116.21:
101

Formula 116.22:
10.049 875
Formula 116.23:
0.049 751
Formula 116.24:
0.049 752
Formula 116.25:
−0.245 91
Formula 116.26:
−0.246 30
Formula 116.27:
102
Formula 116.28:
10.099 504
Formula 116.29:
0.049 507
Formula 116.30:
0.049 507
Formula 116.31:
−0.241 91
Formula 116.32:
−0.242 68
Formula 116.33:
103
Formula 116.34:
10.148 891
Formula 116.35:
0.049 267
Formula 116.36:
0.049 266
Formula 116.37:
−0.239 58
Formula 116.38:
−0.239 16
Formula 116.39:
104
Formula 116.40:
10.198 039
Formula 116.41:
0.049 029
Formula 116.42:
0.049 029
Formula 116.43:
−0.236 91
Formula 116.44:
−0.235 72
Formula 116.45:
105
Formula 116.46:
10.246 950
Formula 116.47:
0.048 795
Formula 116.48:

$0.048\,795$
Formula 116.49:
$-0.236\,66$
Formula 116.50:
$-0.232\,36$
Formula 116.51:
$S(x)$
Formula 116.52:
$f(x)$
Formula 116.53:
$S_3(x)$
Formula 116.54:

$$|f^{(a)}(x) - S_3^{(a)}(x)| = O(h^{4-a}) \qquad (a = 0, 1, 2, 3),$$

Formula 116.55:

$$f^{(a)}(x) \approx S_3^{(a)}(x) \qquad (a = 0, 1, 2, 3). \tag{4.5.6}$$

Formula 116.56:
x
Formula 116.57:
x_i
Formula 116.58:

$$\pi : a = x_0 < x_1 < x_2 < \cdots < x_n = b, \quad x_{k+1} - x_k = h,$$

Formula 116.59:
$S_3(x)$
Formula 116.60:
$S_3'(x_k) = m_k$
Formula 116.61:
$M_i / \times 10^3$
Formula 116.62:
$f''(x_i) / \times 10^3$
Formula 116.63:
10^3
Formula 116.64:
$f''(100) = -1/(4 \cdot 100^{3/2}) = -0.00025$
Formula 116.65:
-0.25000

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Formula 117.1:

$$m_{k-1} + 4m_k + m_{k+1} = 3(y_{k+1} - y_{k-1})/h, \quad k = 1, 2, \cdots, n-1. \tag{4.5.7}$$

Formula 117.2:
$m_0 = y_0', m_n = y_n'$

Formula 117.3:
m_k

Formula 117.4:
$f'(x_k)$

Formula 117.5:
$P(x)$

Formula 117.6:
$f(x)$

Formula 117.7:
$P(x)$

Formula 117.8:
$f(x)$

Formula 117.9:

$$\int_{-h}^{h} f(x)\,\mathrm{d}x \approx A_{-1}f(-h) + A_0 f(0) + A_1 f(h);$$

Formula 117.10:

$$\int_{-2h}^{2h} f(x)\,\mathrm{d}x \approx A_{-1}f(-h) + A_0 f(0) + A_1 f(h);$$

Formula 117.11:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx [f(-1) + 2f(x_1) + 3f(x_2)]/3;$$

Formula 117.12:

$$\int_{0}^{h} f(x)\,\mathrm{d}x \approx h[f(0) + f(h)]/1 + ah^2[f'(0) - f'(h)].$$

Formula 117.13:

$$\int_0^1 \frac{x}{4+x^2}\,\mathrm{d}x, \quad n = 8;$$

Formula 117.14:

$$\int_0^1 \frac{(1-e^{-x})^{\frac{1}{2}}}{x}\,\mathrm{d}x, \quad n = 10;$$

Formula 117.15:
$h[f(0) + f(h)]/1$

Formula 117.16:
$f \equiv 1$

Formula 117.17:
h

Formula 117.18:
$2h$

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Formula 118.1:

$$\int_1^9 \sqrt{x}\,\mathrm{d}x, \quad n = 4;$$

Formula 118.2:

$$\int_0^{\pi/6} \sqrt{4 - \sin^2\varphi}\,\mathrm{d}\varphi, \quad n = 6.$$

Formula 118.3:
$\int_0^1 e^{-x}\,\mathrm{d}x$
Formula 118.4:

$$\int_a^b f(x)\,\mathrm{d}x = (b-a)f(a) + \frac{f'(\eta)}{2}(b-a)^2,$$

Formula 118.5:

$$\int_a^b f(x)\,\mathrm{d}x = (b-a)f(b) - \frac{f'(\eta)}{2}(b-a)^2,$$

Formula 118.6:

$$\int_a^b f(x)\,\mathrm{d}x = (b-a)f\left(\frac{a+b}{2}\right) + \frac{f''(\eta)}{24}(b-a)^3.$$

Formula 118.7:
$n \to \infty$
Formula 118.8:
$\int_a^b f(x)\,\mathrm{d}x$
Formula 118.9:
$\int_a^b f(x)\,\mathrm{d}x$
Formula 118.10:
$[a, b]$
Formula 118.11:
ε
Formula 118.12:
$\frac{2}{\sqrt{\pi}}\int_0^1 e^{-x}\,\mathrm{d}x$
Formula 118.13:
10^{-5}
Formula 118.14:

$$s = a\int_0^{\pi/2} \sqrt{1 - \left(\frac{c}{a}\right)^2 \sin^2\theta}\,\mathrm{d}\theta,$$

Formula 118.15:
a
Formula 118.16:
c
Formula 118.17:
h
Formula 118.18:

H
Formula 118.19:
$R = 6\,371$ km
Formula 118.20:
$a = (2R + H + h)/2, \quad c = (H - h)/2$
Formula 118.21:
$h = 439$ km
Formula 118.22:
$2\,384$ km
Formula 118.23:

$$n \sin \frac{\pi}{n} = \pi - \frac{\pi^3}{3!\, n^2} + \frac{\pi^5}{5!\, n^4} - \cdots ,$$

Formula 118.24:
$n \sin \frac{\pi}{n}$
Formula 118.25:
$n = 3, 6, 12$
Formula 118.26:
π
Formula 118.27:
$\int_1^3 \frac{\mathrm{d}y}{y}$
Formula 118.28:
$f(x) = \frac{1}{(1+x)^2}$
Formula 118.29:
$x = 1.0, 1.1$
Formula 118.30:
1.2
Formula 118.31:
$f(x)$
Formula 118.32:
x
Formula 118.33:
1.0
Formula 118.34:
1.1
Formula 118.35:
1.2
Formula 118.36:
1.3
Formula 118.37:
1.4
Formula 118.38:
$f(x)$
Formula 118.39:
0.250 0
Formula 118.40:
0.226 8
Formula 118.41:
0.206 6
Formula 118.42:
0.189 0

Formula 118.43:
$0.173\,6$
Formula 118.44:
$s = a \int_0^{\pi/2} \cdots \, \mathrm{d}\theta$
Formula 118.45:
s
Formula 118.46:
4
Formula 118.47:
$c = 0$
Formula 118.48:
$\pi a/2$
Formula 118.49:
$2\pi a$